Test Case Documentation (guideline §6.3.3)

Functional test cases for the key requirements, in the §6.3.3 format. The complete list of automated tests is `test-register.md`; this table gives the full detail (requirement, preconditions, input, expected and actual output) for the cases that prove each functional requirement (FR-001 to FR-015, SDS Table 2).

- **Date executed:** 2026-10-28.
- **Where the Actual Output comes from:**
- **Live** = run against the staging environment (`https://staging.eduralms.com`) with `edura-tests/integration` (branch DDP-#66, 197 passed, 1 skipped) or the Playwright suite `edura-tests/e2e` (DDP-#68, 6 passed).
- **CI** = passed in the Backend CI run for the 11 services (778 tests) or in the in-process mock-gateway run of `edura-tests/integration/test_payment_enrollment_flow.py`.
- **Status key:** Pass, Fail, Blocked. A requirement with no automated test is listed in "Requirements without an automated test" instead of being given an invented result.
- Priority: P1 = must pass, P2 = should pass.

FR-001: Authentication and role-based access

Field	TC-FR-01-01	TC-FR-01-02	TC-FR-01-03	TC-FR-01-04
Name	Student logs in with valid credentials	Wrong password is refused	Student cannot use an admin endpoint	Logout revokes the refresh token
Requirement	FR-001	FR-001	FR-001	FR-001
Test file	`integration/test_auth_flow.py`	`integration/test_auth_flow.py`	`integration/test_admin_flow.py`	`integration/test_auth_flow.py`
Test	`test_given_valid_credentials_when_login_then_returns_bearer_token`	`test_given_wrong_password_when_login_then_401`	`test_given_student_when_get_admin_root_then_403`	`test_given_logged_in_session_when_logout_t`
Type / Priority	Integration / P1	Integration / P1	Integration / P1	Integration / P1
Preconditions	A verified student exists	A verified student exists	A verified student is logged in	A student is logged in and holds a refresh token
Input	`POST /api/auth/login` with the correct email and password	`POST /api/auth/login` with a wrong password	the admin root endpoint with the student's bearer token	`POST /api/auth/logout`, then `POST /api/auth/` token
Expected output	HTTP 200; a bearer `access_token`	HTTP 401	HTTP 403	Logout 200; the later refresh returns HTTP 401
Actual output	HTTP 200, bearer token returned	HTTP 401	HTTP 403	Logout 200; refresh 401
Status	Pass (Live)	Pass (Live)	Pass (Live)	Pass (Live)

FR-002: Registration and email verification

Field	TC-FR-02-01	TC-FR-02-02	TC-FR-02-03	TC-FR-02-04
Name	New student registers	Weak password is rejected	Valid verification link verifies the account	Invalid verification token is refuse
Requirement	FR-002	FR-002	FR-002	FR-002
Test file	`integration/test_auth_flow.py`	`integration/test_auth_flow.py`	`integration/test_auth_flow.py`	`integration/test_auth_flow.`
Test	`test_given_new_email_when_register_then_returns_unverified_student`	`test_given_weak_password_when_register_then_422`	`test_given_valid_token_when_verify_then_account_is_verified`	`test_given_garbage_token_wh`
Type / Priority	Integration / P1	Integration / P1	Integration / P1	Integration / P1
Preconditions	The email is not registered	None	An unverified account exists; a token signed for it	None
Input	`POST /api/auth/register` with a new email, a strong password and a profile	`POST /api/auth/register` with a password that breaks the policy	`POST /api/auth/verify-email` with the signed token	`POST /api/auth/verify-email`
Expected output	Account created, not yet verified	HTTP 422	Account marked verified; login now succeeds	HTTP 400
Actual output	Account created, unverified	HTTP 422	Account verified	HTTP 400
Status	Pass (Live)	Pass (Live)	Pass (Live)	Pass (Live)

UI evidence: Playwright `e2e/tests/01-register-and-login.spec.ts` registers through the form, verifies, signs in and is redirected away from `/login` (Pass, Live). A duplicate email returns HTTP 400 (`test_given_existing_email_when_register_then_400_email_exists`, Pass, Live).

FR-003: Content access and upload rules

Field	TC-FR-03-01	TC-FR-03-02	TC-FR-03-03
Name	Admin streams a lesson	Unenrolled student cannot stream	Disallowed file type is refused
Requirement	FR-003	FR-003	FR-003
Test file	`integration/test_content_flow.py`	`integration/test_content_flow.py`	`integration/test_content_flow.py`
Test	`test_given_admin_when_stream_lesson_then_embed_url_built_from_video_id`	`test_given_unenrolled_student_when_stream_then_403`	`test_given_disallowed_mime_when_upload_then_415`
Type / Priority	Integration / P1	Integration / P1	Integration / P2
Preconditions	Admin token	A student not enrolled in the course	Admin token
Input	`GET /api/content/lessons/{id}/stream` with a video id	The same request with the student's token	`POST` upload with a disallowed MIME type
Expected output	HTTP 200; embed URL built from the video id	HTTP 403 (not enrolled)	HTTP 415
Actual output	HTTP 200, embed URL returned	HTTP 403	HTTP 415
Status	Pass (Live)	Pass (Live)	Pass (Live)

FR-004 and FR-005: Assessments and automatic grading

Field	TC-FR-04-01	TC-FR-04-02	TC-FR-05-01	T
Name	Admin adds an MCQ question	Question without a correct answer is rejected	Correct answers give full marks	W
Requirement	FR-004	FR-004	FR-005	F
Test file	`integration/test_assessment_flow.py`	`integration/test_assessment_flow.py`	`integration/test_assessment_flow.py`	`i`
Test	`test_given_assessment_when_admin_adds_mcq_then_correct_answer_taken_from_option`	`test_given_no_correct_answer_when_add_question_then_422`	`test_given_correct_answer_when_submit_then_full_score_and_passed`	`t`
Type / Priority	Integration / P1	Integration / P1	Integration / P1	In
Preconditions	An assessment exists; admin token	An assessment exists; admin token	An enrolled student has an active exam session	A
Input	Add a question with options, one marked correct	Add a question with no option marked correct	Submit the correct option	S
Expected output	Question stored with its correct answer	HTTP 422	Full score; result passed	S
Actual output	Question stored	HTTP 422	Full score, passed	S
Status	Pass (Live)	Pass (Live)	Pass (Live)	P

UI evidence: Playwright `e2e/tests/03-take-exam.spec.ts` enrols, starts an exam, answers every question, submits and sees "Exam Results" with "Marks Obtained" (Pass, Live).

FR-006: Bank-slip (receipt) payment

Field	TC-FR-06-01	TC-FR-06-02
Name	Admin approves an uploaded receipt	Fake PDF is refused
Requirement	FR-006	FR-006
Test file	`payment_service/tests/test_payment_events.py`	`payment_service/tests/test_payment_events.py`
Test	`test_given_uploaded_receipt_when_admin_approves_then_payment_and_review_announced`	`test_given_pdf_and_fake_pdf_when_receipt_uploaded_then_real_one_stored_and_fake_refused`
Type / Priority	Unit / P1	Unit / P2
Preconditions	A receipt is uploaded for a payment	A pending payment
Input	Admin approves the receipt	Upload a file named `.pdf` that is not a PDF
Expected output	Payment confirmed; review event published	Real PDF stored; fake refused
Actual output	As expected	As expected
Status	Pass (CI)	Pass (CI)

FR-007 to FR-009: PayHere payment, webhook and automatic access

Field	TC-FR-07-01	TC-FR-08-01	TC-FR-08-02	TC-FR-09-01
Name	Paid course opens checkout	Webhook with an invalid signature is rejected	Duplicate webhook is idempotent	Success wel
Requirement	FR-007	FR-008	FR-008	FR-009
Test file	`e2e/tests/02-browse-and-enrol.spec.ts`	`integration/test_payment_enrollment_flow.py`	`integration/test_payment_enrollment_flow.py`	`integratio`
Test	`test_given_a_paid_course_when_the_student_chooses_to_purchase_then_checkout_opens_with_the_course_price`	`test_ac3_invalid_hmac_blocks_enrollment`	`test_ac4_duplicate_webhook_is_idempotent`	`test_ac1_w`
Type / Priority	E2E / P1	Integration / P1	Integration / P1	Integration /
Preconditions	A signed-in student; paid course 10 (LKR 2,500)	A pending payment with an order id	A payment already confirmed by one webhook	A pending pa
Input	Click Buy on the course page	`POST /webhook` with a wrong signature	The same valid webhook sent twice	`POST /webh` success stat
Expected output	Browser at `/payment?type=course&id=10` showing the price	HTTP 400; no enrolment created	Both answered HTTP 200 SUCCESS; exactly one payment record	HTTP 200 S enrolment A
Actual output	Checkout opened with LKR 2,500	HTTP 400; enrolment lookup 404	Both 200 SUCCESS; one record	Payment SU
Status	Pass (Live)	Pass (CI, mock gateway)	Pass (CI, mock gateway)	Pass (CI, mo

The webhook cases ran only against the mock gateway. Running them against staging needs the staging PayHere credentials, which were not available (listed in the coverage report gaps).

FR-010: Progress tracking

Field	TC-FR-10-01	TC-FR-10-02
Name	Watching a lesson is recorded	Dashboard count increases
Requirement	FR-010	FR-010
Test file	`integration/test_progress_flow.py`	`integration/test_progress_flow.py`
Test	`test_given_student_when_mark_lesson_watched_then_progress_recorded`	`test_given_watched_lesson_when_get_dashboard_then_watched_count_increases`
Type / Priority	Integration / P1	Integration / P1
Preconditions	An enrolled student	A student who has marked a lesson watched
Input	Mark a lesson watched	`GET` the progress dashboard before and after
Expected output	Progress recorded	Watched count is higher after
Actual output	Progress recorded	Count increased
Status	Pass (Live)	Pass (Live)

FR-013 and FR-014: Anti-cheating and gamification

Field	TC-FR-13-01	TC-FR-14-01
Name	Violation during an exam is recorded	Public badge catalogue is available
Requirement	FR-013	FR-014
Test file	`integration/test_assessment_flow.py`	`integration/test_assessment_flow.py`
Test	`test_given_session_when_record_violation_then_204`	`test_given_anonymous_visitor_when_list_badges_then_catalogue_returned`
Type / Priority	Integration / P2	Integration / P2
Preconditions	An active exam session	None
Input	Record a violation for the session	`GET` the badge list without a token
Expected output	HTTP 204	The catalogue of badges
Actual output	HTTP 204	Catalogue returned
Status	Pass (Live)	Pass (Live)

FR-012: Analytics

Field	TC-FR-12-01
Name	Admin reads platform analytics; a student cannot
Requirement	FR-012
Test file	`integration/test_notification_analytics_flow.py`
Test	`test_given_admin_when_get_admin_stats_then_200`, `test_given_student_when_get_admin_stats_then_403`
Type / Priority	Integration / P2
Preconditions	An admin token and a student token
Input	`GET` the admin statistics endpoint with each token
Expected output	Admin 200; student 403
Actual output	Admin 200; student 403
Status	Pass (Live)

FR-015: Completion certificates

Field	TC-FR-15-01	TC-FR-15-02	TC-FR-15-03
Name	Completed course issues a certificate	Failed assessment blocks the certificate	Certificate is not issued twice
Requirement	FR-015 (BR-004)	FR-015 (BR-004)	FR-015
Test file	`progress_service/tests/test_progress.py`	`progress_service/tests/test_progress_service.py`	`progress_service/tests/test_pr`
Test	`test_given_completed_course_when_certificate_issued_then_record_created_and_event_published`	`test_given_failed_assessment_when_course_completed_then_certificate_is_not_issued`	`test_given_issued_certificate_`
Type / Priority	Unit / P2	Unit / P2	Unit / P2
Preconditions	A student watched 100% of the course and passed every assessment	100% watched, one assessment failed	A certificate already exists for the stu
Input	Completion is recorded	Completion is recorded	Completion is recorded again
Expected output	A certificate record is created and `certificate.issued` is published	No certificate	No second certificate
Actual output	As expected	As expected	As expected
Status	Pass (CI)	Pass (CI)	Pass (CI)

Partly implemented. The eligibility rule, the database record and the `certificate.issued` event work and are tested. The record stores only a placeholder link (`/certificates/<student>_<course>.pdf`): no PDF is generated, nothing stores it, and the frontend has no page to view or download a certificate. The requirement says the certificate is "generated and issued", so it is only partly met.

Requirements not fully covered

Requirement	Position
FR-011 Gemini digital assistant	Not implemented; out of scope for this release (SRS 3.3 excludes advanced AI features). The chat in `admin_service` is the human support chat. No test case is needed.
FR-012 Excel export	**Not implemented.** The backend has no Excel (xlsx) or analytics CSV export. Only the progress report in PDF is built (`analytics_service/report.py`, `POST /students/{user_id}/send-report`, emailed to the student) and it is unit-tested (`analytics_service/tests/test_progress.py`). The PDF half of FR-012 is met; the Excel half is not.
FR-015 Certificate PDF	See above: record and event only, no generated PDF and no frontend view.

These gaps belong in the final report under "Limitations & future work". List FR-011, FR-012 (Excel) and FR-015 (PDF) as known deviations from the SRS.